

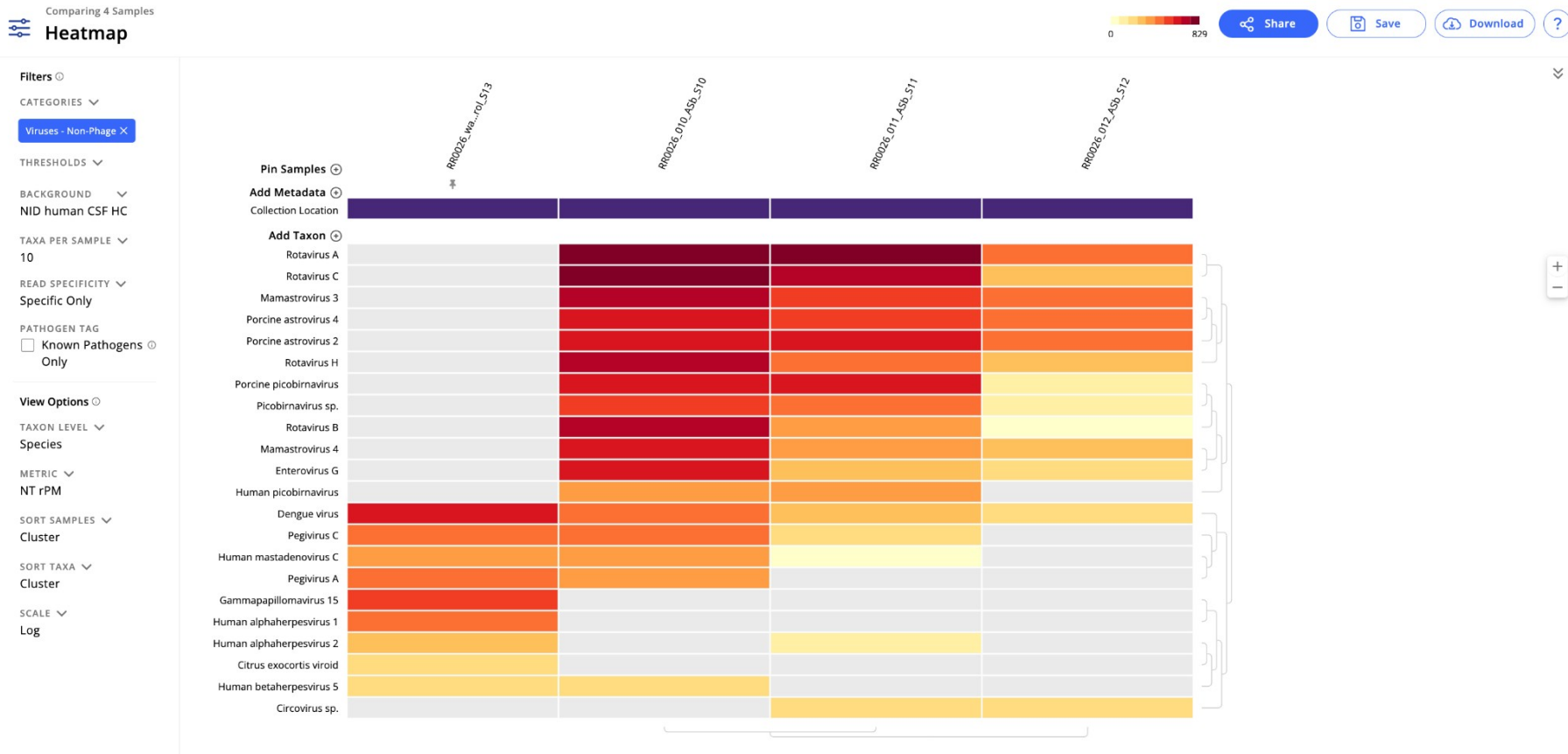
Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis



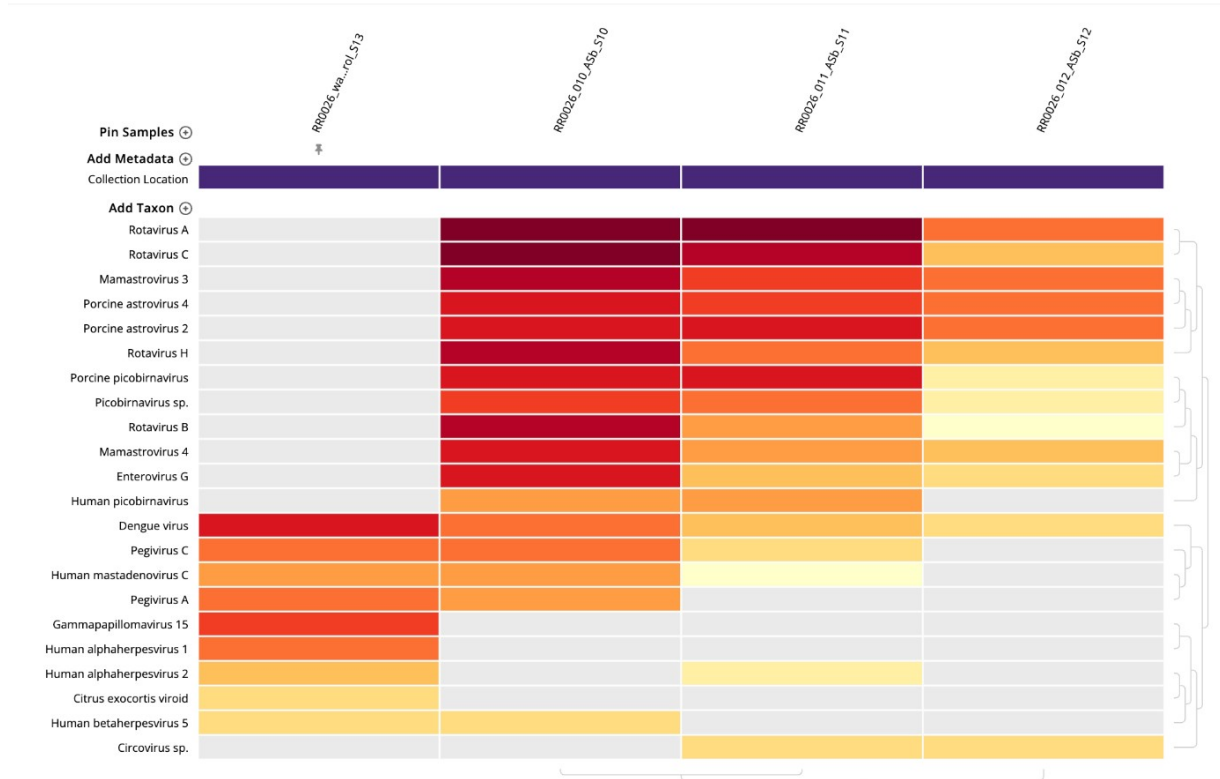
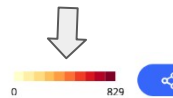
Heatmaps

Heatmap: Enables you to visually compare detected taxa across samples and quickly spot microbes of interest



Taxa

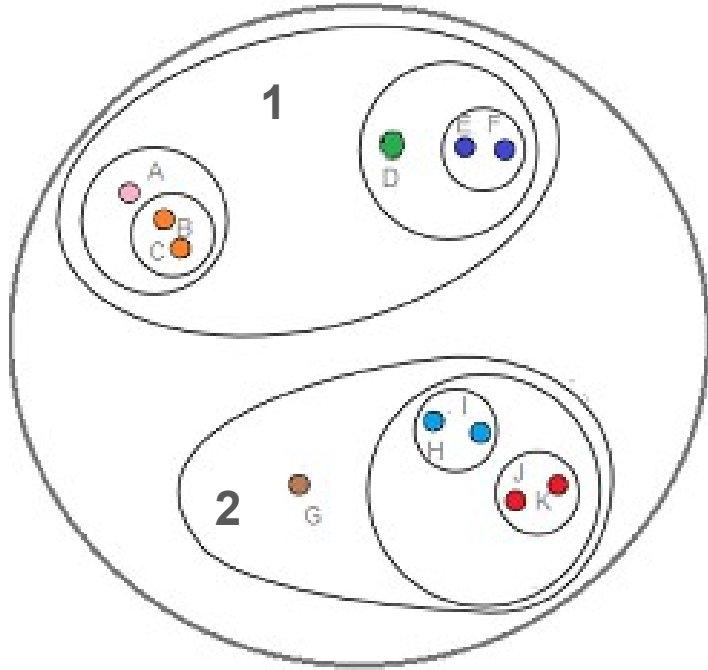
Scale
(default: NT rPM)



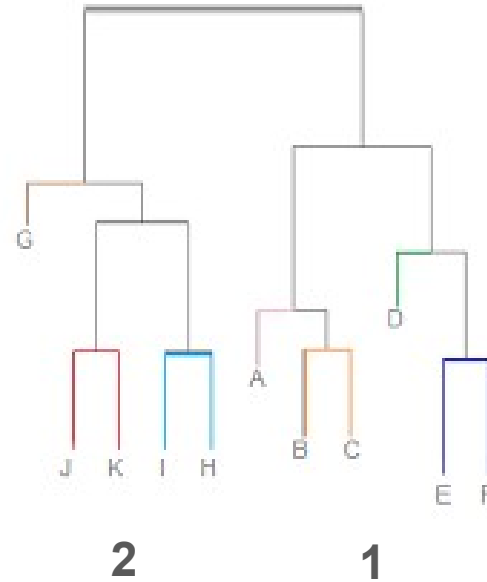
Taxa and samples are organized through hierarchical clustering



Clusters are represented using a dendrogram



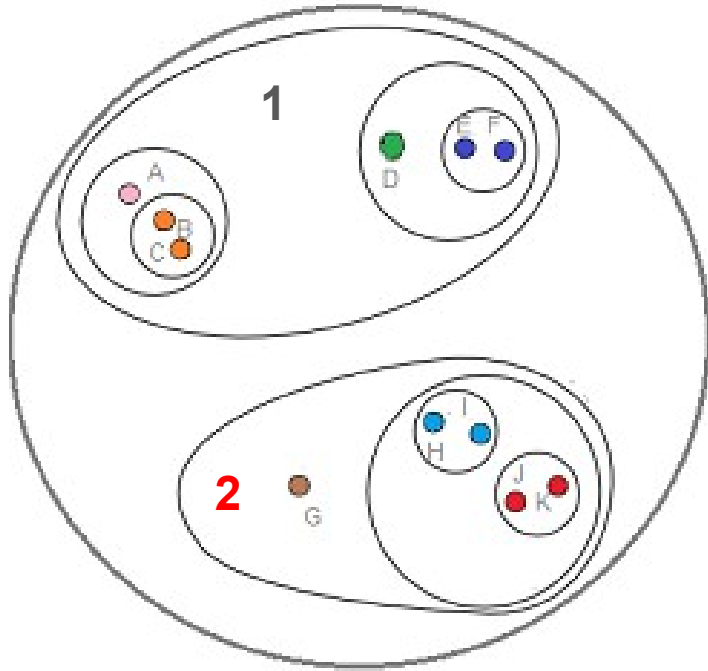
Clusters: Groups that include similar samples based on co-occurring taxa and their relative abundance



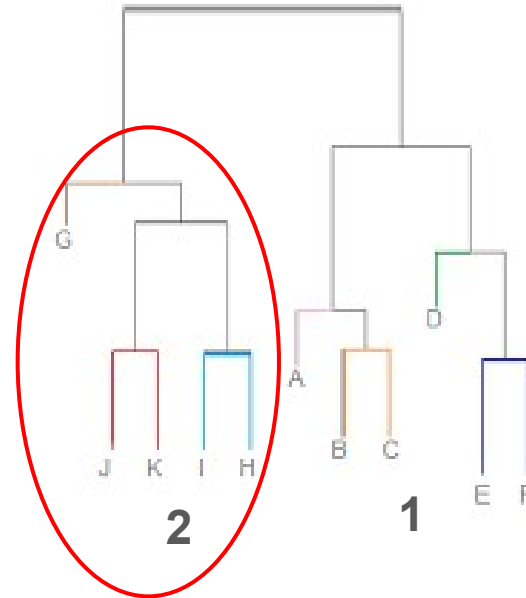
Dendrogram: Tree representation of the data (clusters = clades)

[Image source](#)

Clusters are represented using a dendrogram



Clusters: Groups that include similar samples based on co-occurring taxa and their relative abundance



Dendrogram: Tree representation of the data (clusters = clades)

[Image source](#)

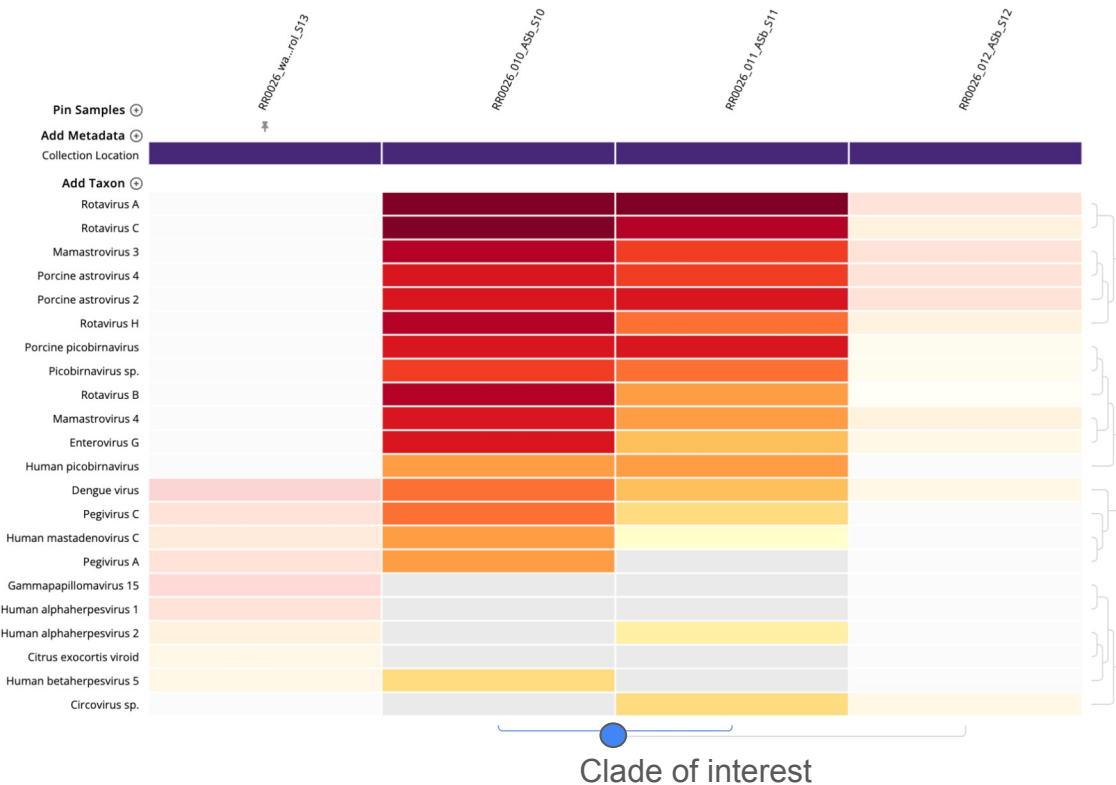
Taxa and samples are organized through hierarchical clustering



Taxa that are in a clade (or cluster) are more likely to appear together across samples.

Samples that are clustered together have similar taxa.

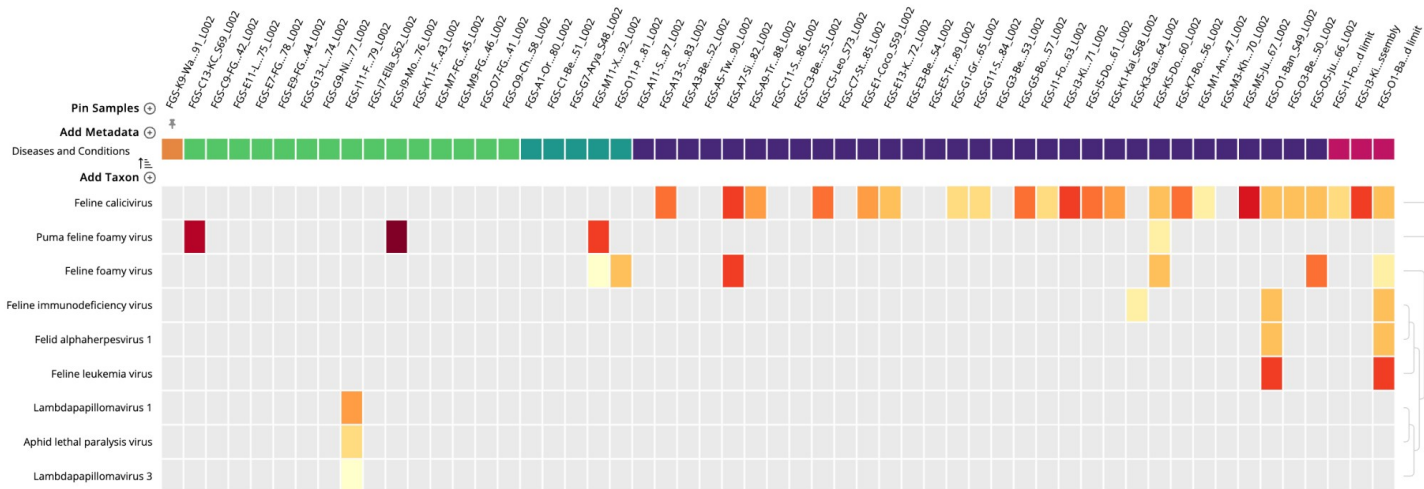
Identify clusters of interest



You can override **sample** clustering to organize heatmap

Pin samples to keep samples of interest on the left

Add metadata and sort samples to identify patterns or microbes of interest



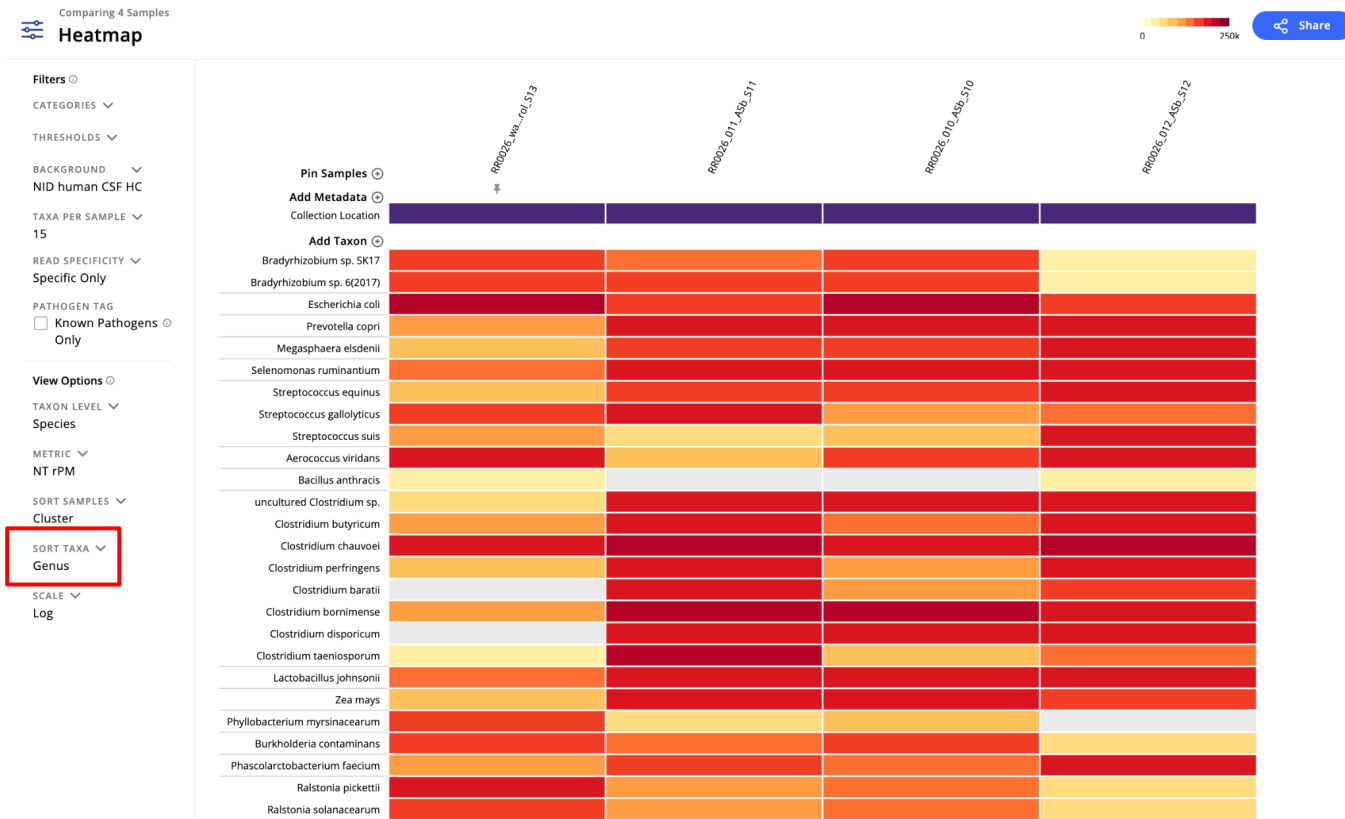
Metadata legend:

- Orange - water control
- Green - healthy cats
- Blue - diseased cats

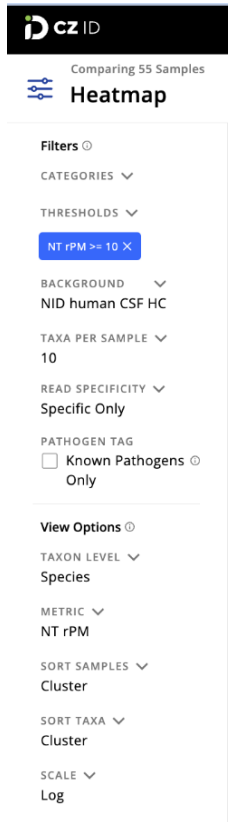
[Feline gingivostomatitis](#)

You can override clustering of **taxa** to organize heatmap

Sorting taxa by
genus may help
when evaluating
contaminants



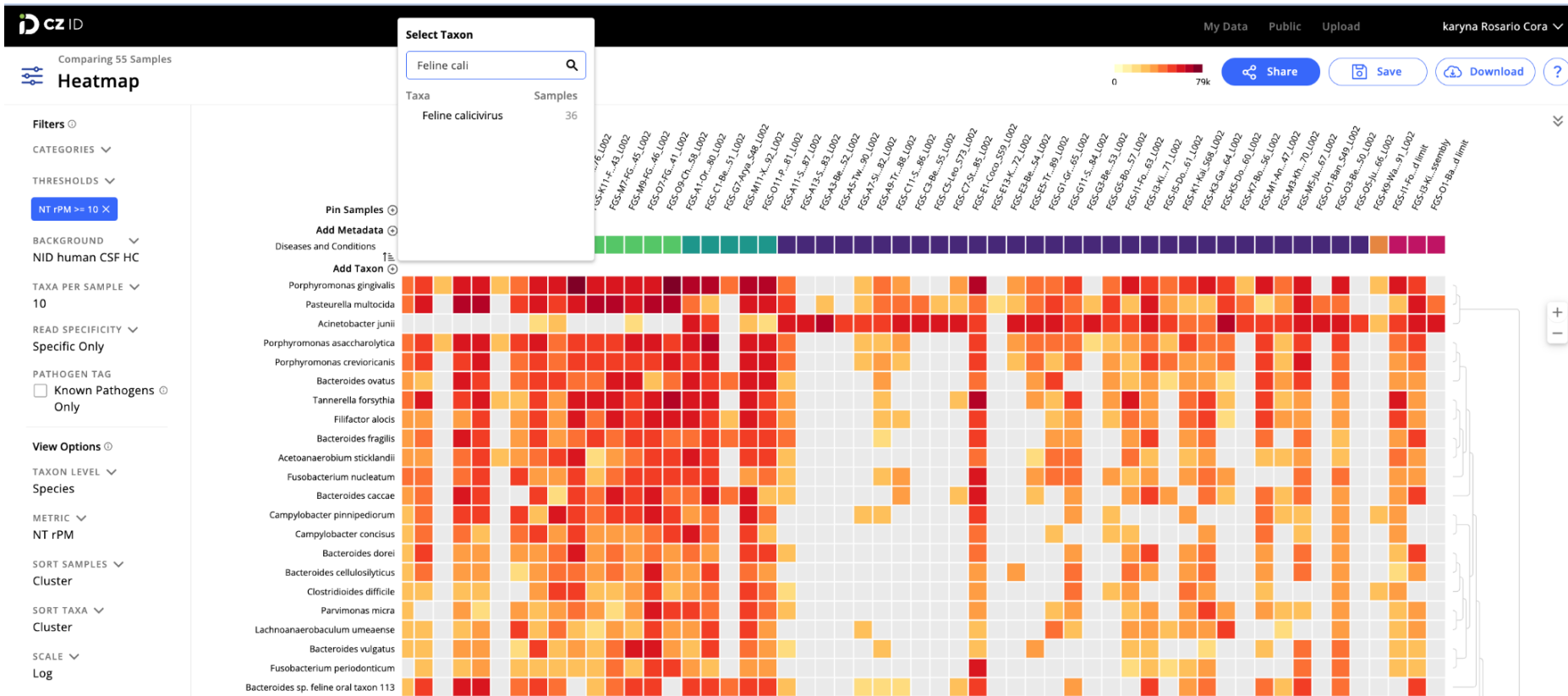
Use “Filters” and “View Options” to explore and reduce noise



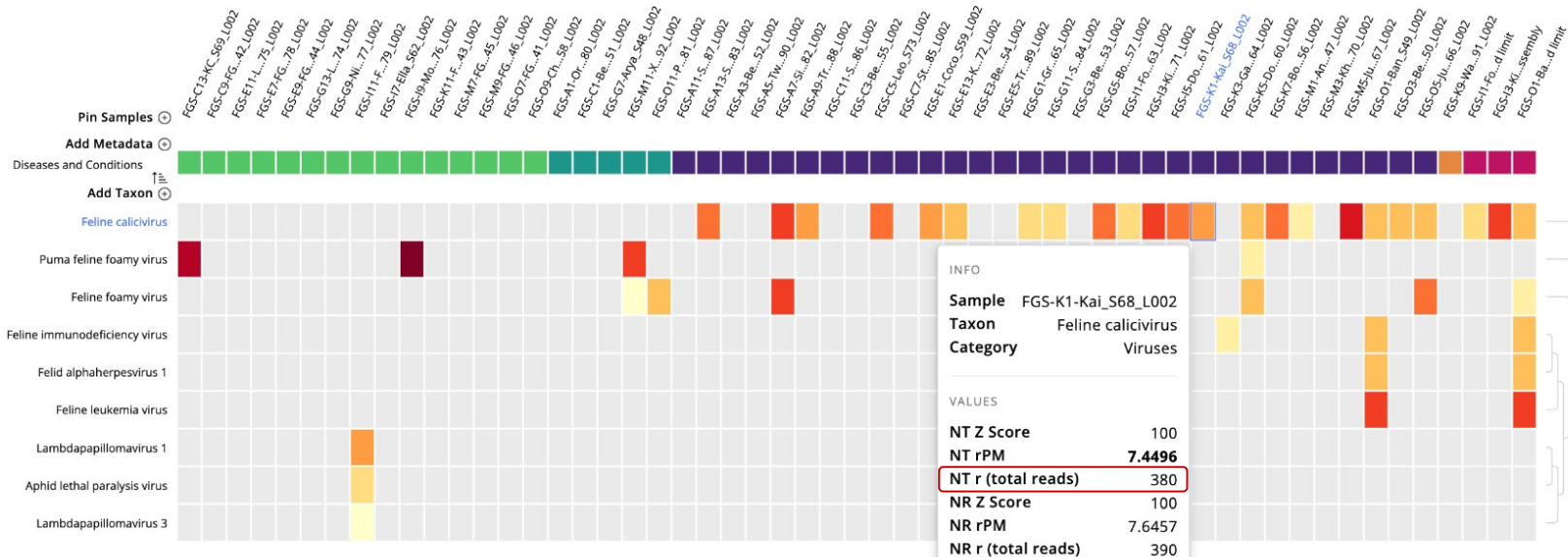
The screenshot shows the CZ ID Heatmap interface. At the top, it says "Comparing 55 Samples" and "Heatmap". Below this is a sidebar with several sections: "Filters" (with sub-sections for Categories, Thresholds, Background, Taxa per Sample, Read Specificity, and Pathogen Tag), and "View Options" (with sub-sections for Taxon Level, Metric, Sort Samples, Sort Taxa, and Scale). The "Filters" section is currently expanded, showing "NT rPM >= 10 X" under Thresholds, "NID human CSF HC" under Background, "10" under Taxa per Sample, "Specific Only" under Read Specificity, and "Known Pathogens Only" under Pathogen Tag. The "View Options" section is also expanded, showing "Species" under Taxon Level, "NT rPM" under Metric, "Cluster" under Sort Samples, "Cluster" under Sort Taxa, and "Log" under Scale.

- Filters
 - You can filter by organism category and specific metric thresholds similar to what you use in the Sample Report.
- Background
 - Search and select the appropriate background to view correct Z-scores
- Taxa per Sample
 - Use this to increase or reduce the number of top taxa per sample used to create the heatmap
- Read Specificity
 - Use “Specific only” to see calls at the species level only. Use “All” to see both species calls and unclassified taxa that don’t have complete taxon lineages
- Pathogen Tag
 - Select to only view known pathogens (based on [CZ ID’s Pathogen List](#))
- Taxon Level
 - Use to change the taxonomic resolution (species or genus level). Viewing the heatmap at the genus level helps to explore patterns at a broad level
- Metric
 - Use to change the metric used to create the heatmap (NT rPM by default)
- Sort Samples
 - You can choose to sort samples alphabetically (this would override sample clustering)
- Scale
 - You can choose between a logarithmic (Log, default) or linear (Lin) scale

Use “Add Taxon” to view low abundance taxa of interest



Hover over a cell to see additional metrics



Let's make a heatmap!

Current Activity: CZ ID Training Reference Tables & Worksheets_Illumina (pg 14-15)

Next Slides (Optional Module): DL9-I_CZ ID AMR Module

Next Slides (Optional Module): DL10-I_Building Viral Consensus Genomes